



Database Architecture

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Introduction

- ▶ **Staged** database design process
 - distinguished between logical database model & physical database model
- ▶ **History**
 - 1975, Similar architecture was developed and published by NSIC (ANSI/X3/SPARC)

Three level of Abstraction

(Three-Level Architecture or
Three Schemas or
Three Models)

Introduction

- ▶ **Model** refer to the permanent model of structure, not data stored

The basic **building blocks** of all the data models are **entities, attributes, relationships and constraints** that are determined on the basis of **business rules** of an organization.

- ▶ **Database model:**

collection of tools

- ▶ **Diagrams**

- ▶ **Specialized** vocabulary to describe structure of database

Database Languages

Introduction

► Data Definition Language (DDL)

Allows the DBA or user to describe and name entities, attributes, and relationships required for the application

plus any associated integrity and security constraints.

E.g. Create Table ...

Database Languages

Introduction

- ▶ **Data Manipulation Language (DML)**
Provides basic data manipulation operations on data held in the database.
Update statement in SQL...

- ▶ **Data Control Language (DCL)**

Purpose of Three-Level Architecture

► User's view

Requirements of each user varies.

All users should be able to access same data.

Users should not need to know physical database storage details.

Purpose of Three-Level Architecture

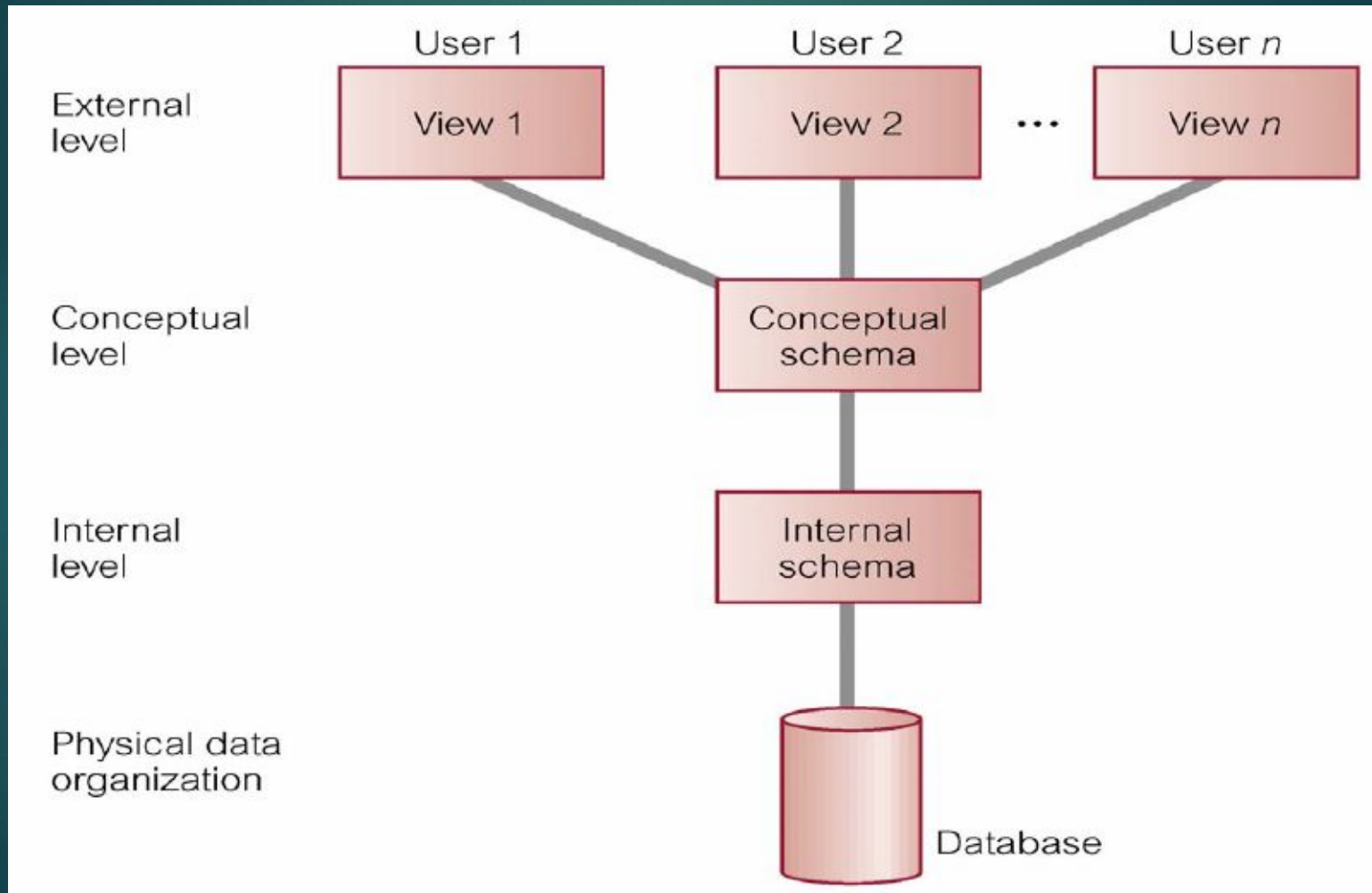
► DBA's view

DBA should be able to change database structures without affecting the users' views.

Internal structure of database should be unaffected by changes to physical aspects of storage.

DBA should be able to change conceptual structure of database without affecting all users.

ANSI-SPARC Three-Level Architecture



External view

- ▶ User's point of view
- ▶ External level consists of many different external views
- ▶ Each external view may have different representations
- ▶ External views are described in External schemas which are written in the Data Definition Language
 - Each user's schema provides complete description of user's view
- ▶ External schemas are used to create GUI

External view

User only interact with certain entities

User will be unaware of other entities, attributes and relationships

Conceptual Model

- ▶ DBA's point of view
- ▶ Middle level that contains entire information of the database
- ▶ Community view of the database.
- ▶ Complete model of working organization
- ▶ Complete logical model
 - Entities, attributes, relationships, constraints on data, semantic information, security and integrity
- ▶ Conceptual model supports external views

Conceptual Model

- ▶ **Conceptual** model is relatively constant
 - DBA is responsible to design it
 - Conceptual model can be expanded
- ▶ **It** is written in DDL, stored in data dictionary and compiled by DBMS

Physical Model

- ▶ Describes how the data is stored in the database.
- ▶ Physical implementation of database
 - Includes data structures and file organization
 - Works with OS
 - ▶ To lay out data on the storage devices
 - ▶ Build indexes
 - ▶ Set the pointers
- ▶ Simply, it is managed by OS but under the instruction of DBM
- ▶ Mapping between logical and physical model
 - Stores in data dictionary

Differences between Three Levels of ANSI-SPARC Architecture

External view 1

sNo	fName	lName	age	salary
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External view 2

staffNo	lName	branchNo
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Conceptual level

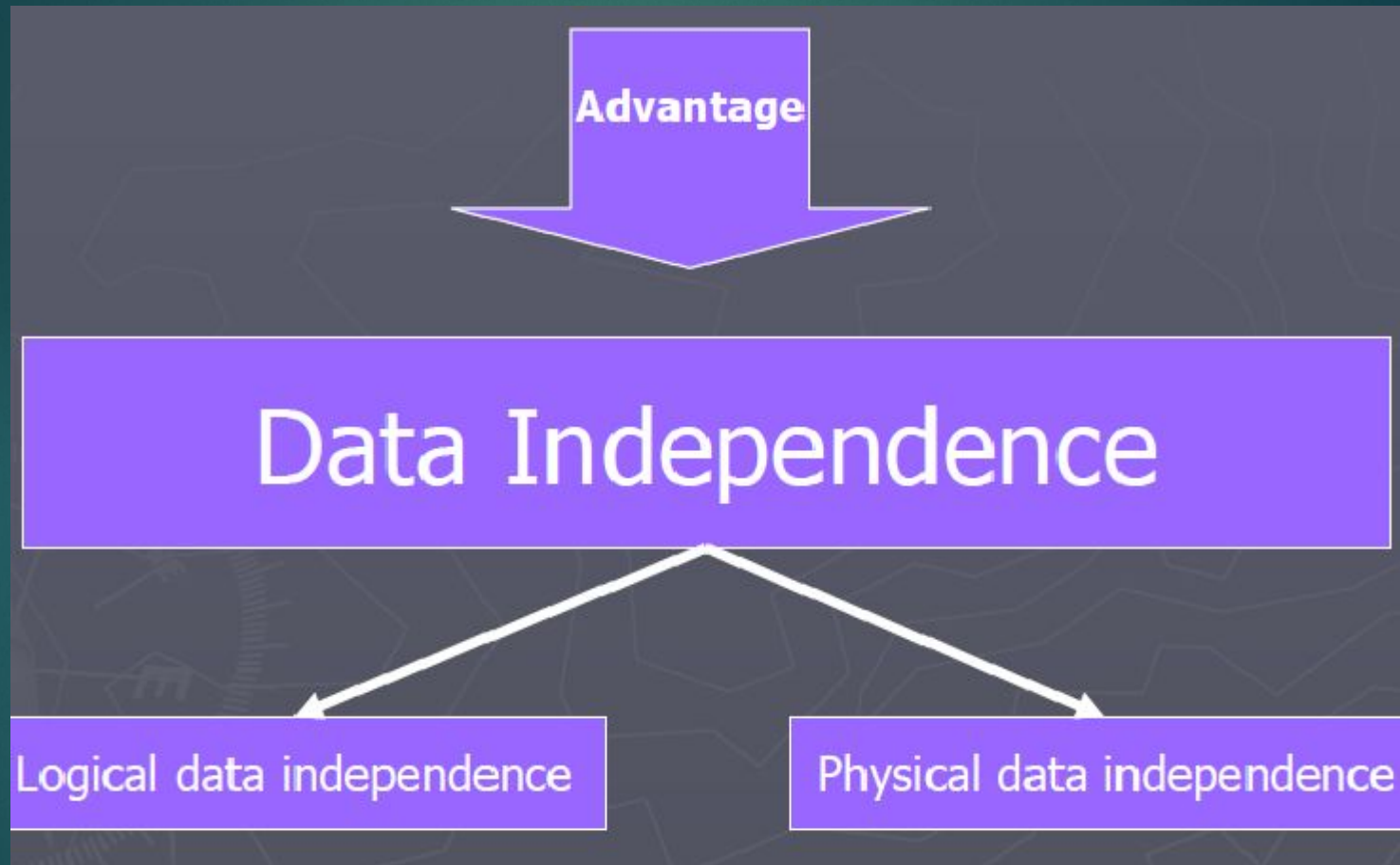
staffNo	fName	lName	DOB	salary	branchNo
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Internal level

```
struct STAFF {  
    int staffNo;  
    int branchNo;  
    char fName [15];  
    char lName [15];  
    struct date dateOf Birth;  
    float salary;  
    struct STAFF *next;  
};  
index staffNo; index branchNo;
```

/* pointer to next Staff record */
/* define indexes for staff */

Three Level Architecture



Data Independence

► Logical Data Independence

Refers to immunity of external schemas to changes in conceptual schema.

Conceptual schema changes (e.g. addition/removal of records)

Should not require changes to external schema or rewrites of application programs.

Data Independence

► Physical Data Independence

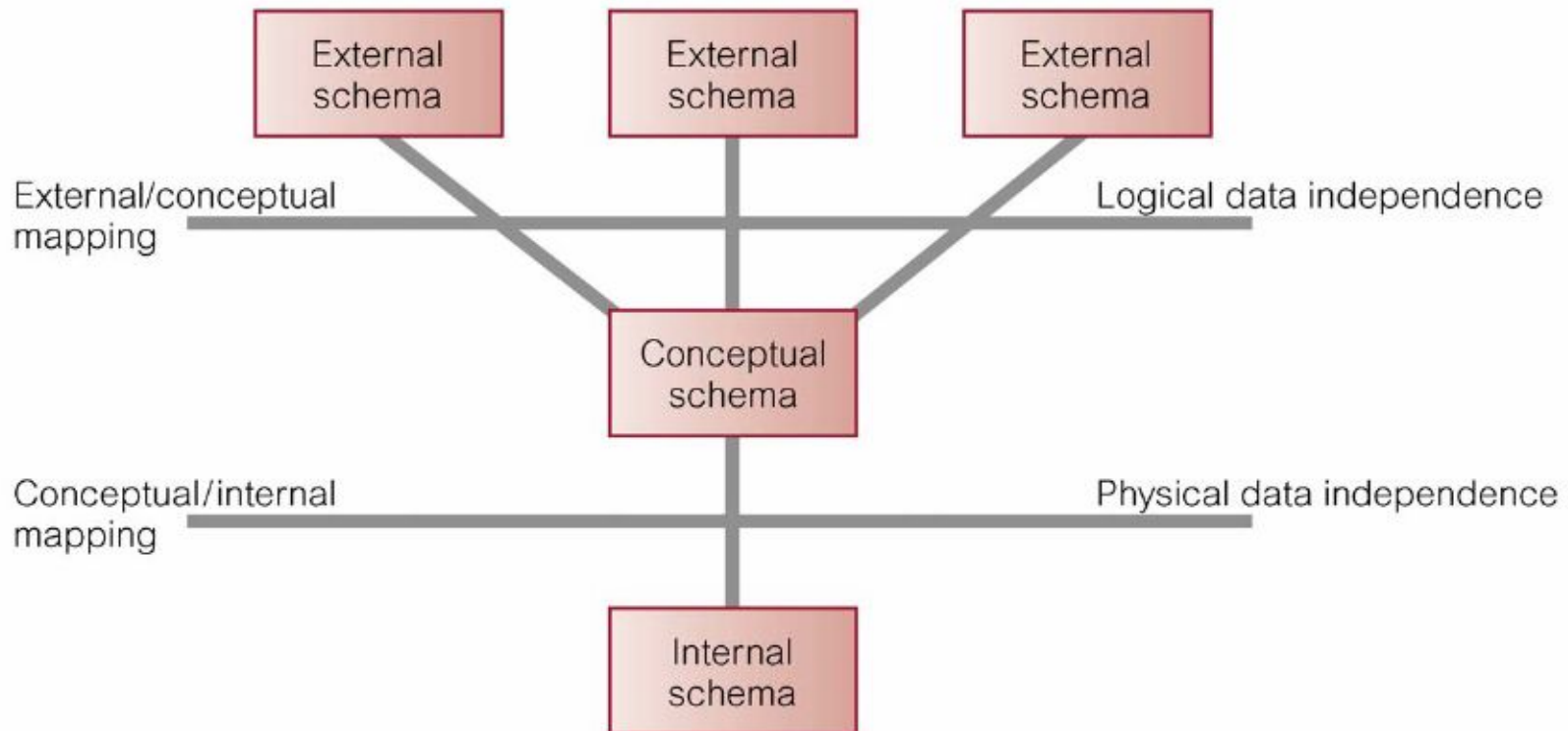
Refers to immunity of conceptual schema to changes in the internal schema.

Internal schema changes (e.g. using different file organizations, storage structures/devices)

► If someone changes his HDD from FAT32 to NTFS!

Should not require change to conceptual or external schemas.

Data Independence and the ANSI-SPARC Three-Level Architecture



Logical Data Models

► Semantics Data Models

The Entity Relationship Model

► Record-Based Models

The Relational Model

The Hierarchical Model

The Network Model

Semantics Data Models

- ▶ To define conceptual and external level of data
- ▶ The most important and power-full model is

The Entity-Relationship Model

Introduced by Chen in mid 1970's
Graphical method